ECE 487
MIDTERM
SOLUTIONS

1. TAU CETI 11 LIGHT YEARS = 1.04×10^{17} m ! NOISE TEMP 38 K 1 MW

S-BAND RADAR $\lambda = 0.12$ m $G = 73$ dB $= 4\pi \frac{A_{eff}}{\lambda^2} \Rightarrow A_{eff} = 2.29 \times 10^{4}$ m^2

AT TAU CETI, $P_{inc} = P_t \frac{G}{4\pi r^2}$ $P_{rec} = P_{inc} \cdot A_{eff} = 3.36 \times 10^{-18}$ W

(ASSUMES FULL POWER TX)

NOISE POWER $= k_B B T = 6.62 \times 10^{-22} B$ W $T_1 + T_2$ 10K BACKGROUND

WANT S/N = 10, $B \cong 500$ Hz SEND 5×8 = 40 BITS PER WORD

→ TAKES $\frac{40}{500} = 0.08$ s

RECEIVED POWER IS 1 MW × 0.08 s = 0.022 KW-HOURS

AT \$.15 / KW-H, A WORD OF DATA COSTS ~ $\frac{1}{3}$ ¢

NOTE: COST DOES NOT DEPEND ON TRANSMIT POWER

2 $$|F(\psi)|^2 = \left| \frac{\sin\{\frac{N+1}{2}(u+u_0)\}}{\sin\{\frac{1}{2}(u+u_0)\}} \right|^2$$ $u = kd\cos\psi$ $N+1$ ELEM.

$u_0 = \alpha d$

NOT NECESSARILY NORMALIZED TO 1 !

SAY $d = 0.4\lambda$; $kd = 0.8\pi$

SUPPOSE $\alpha d = -kd - \frac{x}{N+1}$

SO $u + u_0 = 0.8\pi(\cos\psi - 1) - \frac{x}{N+1}$

$$D = \frac{|F(\psi=0)|^2}{\langle |F(\psi)|^2 \rangle} = \frac{|F|^2|_{\psi=0}}{\frac{1}{4\pi}\int_{0-\pi}\int_{0-2\pi} |F|^2 \sin\psi \, d\psi \, d\phi}$$

$$= 2\,\frac{|F|^2|_{\psi=0}}{\int_0^\pi |F|^2 \sin\psi \, d\psi}$$

CALCULATE D NUMERICALLY

SEE ATTACHED FOR FIGURES, RESULTS, HPBW, ETC.

15 a)

	$D(x=0)$	x_{OPT}	$D(x_{opt})$
5	7.4	1.94	10.6
10	15.3	2.46	24.4
20	31.3	2.71	53

SEE FIGURE

CONVENTIONAL → TENDS TO 2.44

ENHANCED DIR

10 b) SEE FIGURE patterns

c) SEE FIGURE

INCLUDES SIDELOBE EFFECTS

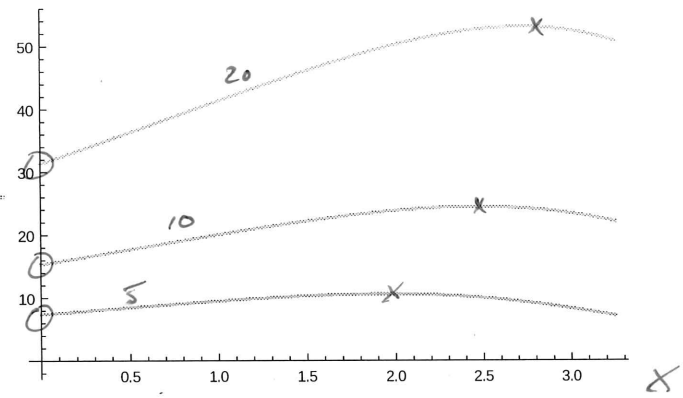
10 ESTIMATE $D = \frac{4\pi}{(HPBW)^2_{RAD}}$ $D = \frac{26{,}000}{(HPBW)^2_{DEG}}$ $D = 4.73\,\frac{L}{\lambda}$

CONVENTIONAL	28 not bad	17.7	37 SO-SO
ENHANCED DIR	80	47.9 not bad	

```
In[1]:= Clear[eps, f2, dir];

In[2]:= eps = 0.0001; (* need this to prevent division by zero problems *)

In[3]:=
        (* below is |f(psi)|^2 definition - note that it may not be normzlied to 1 *)

In[4]:= f2[psi_, x_, n_] := (
          arg = 0.8 Pi (Cos[psi] - 1) - x / n;
          (Sin[0.5 n arg] / (n Sin[0.5 arg]))^2
        );

In[5]:=
        (* here is the definition of directivity
         vs. phase offset correction x and no. array elements n *)

In[6]:= dir[x_, n_] := 2 f2[eps, x, n] / NIntegrate[f2[psi, x, n] Sin[psi], {psi, eps, Pi - eps}];

In[7]:=
        (* plot directivity on linear scale for different sized arrays *)

In[8]:= Plot[{dir[x, 5], dir[x, 10], dir[x, 20]}, {x, 0, 3.25}]
```

D

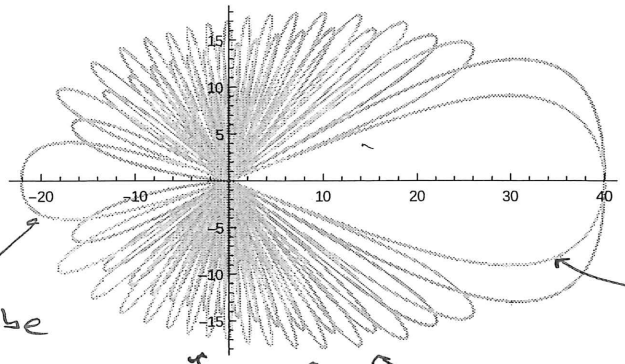
Out[8]=




```
In[9]:= (* compute value of x giving maximum directivity for 20-element array *)

In[10]:= FindMaximum[dir[x, 20], {x, 3}]

         General::stop : Further output of NIntegrate::inum will be suppressed during this calculation. >>

Out[10]= {52.9834, {x → 2.71318}}

In[11]:= (* result is more like e than Pi *)

         (* finally, plot radiation pattern for 20 - element arrays -
          be careful to normalize both plots to same maximum value *)

         f2db[psi_, x_] := Max[10 Log [10, f2[psi, x, 20] / f2[eps, x, 20]] + 40, 0];
```

In[12]:=
```
PolarPlot[{f2db[psi, 0], f2db[psi, 2.713]}, {psi, -Pi, Pi}]
```

Out[12]=



20- ELEMENTS

* ENHANCED DIRECTIVITY

larger backlobe *

narrower main beam *

higher sidelobes *

In[13]:=
```
(* calculate half-power points *)
```

In[14]:=
```
FindRoot[f2[x, 0, 20 ] == f2[eps, 0, 20] / 2.0, {x, 0.2}]
```

Out[14]= {x → 0.334508}

$$HPBW = \frac{180}{\pi} \times 2 \times 0.3345 = 38.3^\circ$$

In[15]:=
```
FindRoot[f2[x, 2.713, 20] == f2[eps, 2.713, 20] / 2.0, {x, 0.1}]
```

{x → 0.203411}

$$HPBW = 23.3^\circ$$

UNIVERSALIZE :

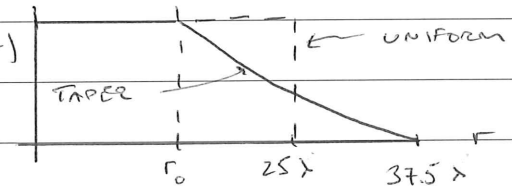
3. a) $F(\theta) \propto \int E_{tan}(r')\, J_0(kr' \sin\theta)\, r'\, dr'$ WRITE $\lambda = 1$ TO SIMPLIFY ALGEBRA

- RADIATION PATTERN VS. THETA PLOTTED FOR UNIFORM E

5 - FIND $\theta_{1/2} = 0.0103$ RAD, HPBW $\cong 1.18^\circ = 0.0206 \cong \underline{1.02 \frac{\lambda}{D}}$

5 - FIND SIDELOBE LEVEL $\sim \underline{-17.57 \text{ dB}}$

b)



5 I FIND $r_0 \sim \underline{14\lambda}$ FOR SAME HPBW AS UNIFORM CASE (~ 9.3 m)

5 - NOW, SIDELOBE LEVEL $\sim \underline{-26.27 \text{ dB}}$

COMPARED TO MAIN LOBE

c) $D = \frac{P_{max}}{P_{avg}} = 4\pi \frac{P_{max}}{P_{tot}} \sim \underline{2.47 \times 10^4} \rightarrow \underline{43.9 \text{ dB}}$

UNIFORM CASE

$= 4\pi \frac{P_{max}}{P_{tot}} \cong \underline{2.79 \times 10^4} \rightarrow \underline{44.5 \text{ dB}}$ TAPER CASE

5 $\epsilon_{UNIFORM} = \frac{A_{eff}}{A_{PHYS}} = \frac{\lambda^2 D/4\pi}{\pi (D/2)^2} = \frac{\lambda^2 D/4\pi}{\pi (25\lambda)^2} = 1.0$ (CHECKS)

DIAMETER

5 $\epsilon_{TAPER} = \frac{A_{eff}}{A_{PHYS}} = \frac{\lambda^2 D/4\pi}{\pi (37.5\lambda)^2} = \underline{0.5}$

d) $\epsilon = \frac{\langle E \rangle^2}{\langle |E|^2 \rangle} \cong 1$ FOR UNIFORM EXCITATION (obvious)

5 $= \frac{\left(\int r\,dr\,d\phi\, E\right)^2 / a^2}{\int r\,dr\,d\phi\, E^2 / a} = \frac{1}{a} \frac{\left(\int r\,dr\,d\phi\, E\right)^2}{\int r\,dr\,d\phi\, E^2}$

$= \frac{2\pi}{a} \frac{\left(\int r\,dr\, E(r)\right)^2}{\int r\,dr\, E(r)^2} = \frac{2}{(37.5)^2} \frac{(253.68)^2}{182.2} = \underline{0.5}$

(CHECKS)

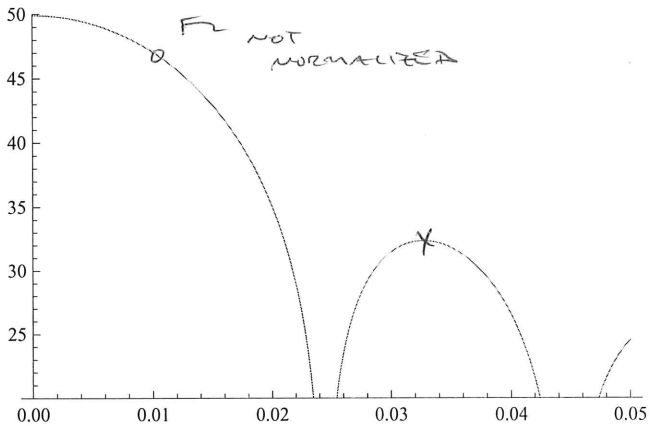
```
In[48]:= g[theta_, r_] := BesselJ[0, 2 Pi r Sin[theta]];

In[49]:= f[theta_] := NIntegrate[g[theta, r] r, {r, 0, 25}];
```

} UNIFORM

```
In[50]:= p[theta_] := 10 Log[10.0, f[theta]^2];

In[51]:= Plot[p[theta], {theta, 0, 0.05}, PlotRange → {20, 50}]
```

dB SCALE

Out[51]=




```
In[52]:= FindRoot[p[a] - (p[0] - 3), {a, 0.0, 0.02}]

Out[52]= {a → 0.0102736}
```

← $\theta_{1/2}$

```
In[53]:= FindRoot[D[p[a], a], {a, 0.03, 0.04}]
```

SIDELOBE MAX

```
Out[53]= {a → 0.0327002}

In[54]:= p[0] - p[0.0327]

Out[54]= 17.5701
```

← 1ST SIDELOBE LEVEL

```
In[55]:= NIntegrate[f[theta]^2 Sin[theta], {theta, 0, Pi / 2}] * 2 Pi

Out[55]= 49.746
```

$\propto P_{tot}$ = α integral

```
In[56]:= f[0]^2

Out[56]= 97656.3
```

$\propto P_{max}$

```
In[95]:= h[r_, r0_, r1_] := (
           If[r < r0, 1, ((r1 - r) / (r1 - r0))^2]
         )
```

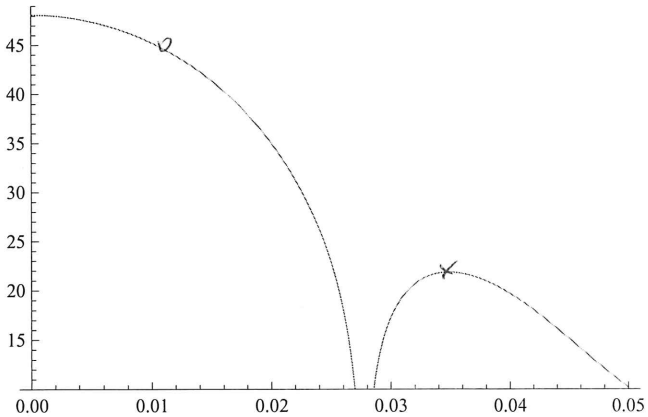
} QUADRATIC TAPER

```
In[65]:= Clear[f]; Clear[p];

In[66]:= f[theta_, r0_] := NIntegrate[g[theta, r] r h[r, r0, 37.5], {r, 0, 37.5}];

In[67]:= p[theta_, r0_] := 10 Log[10.0, f[theta, r0]^2];

In[119]:=
         r0 = 14;
```

} my r_0, in λ

In[122]:=
```
Plot[p[theta, r0], {theta, 0, 0.05}, PlotRange → {10, 49}]
```

Out[122]=



In[120]:=
```
FindRoot[p[a, r0] - (p[0, r0] - 3), {a, 0.0, 0.02}]
```

Out[120]= {a → 0.0102603} ← $\theta_{1/2}$

In[123]:=
```
FindRoot[D[p[a, r0], a], {a, 0.03, 0.04}]
```

Out[123]= {a → 0.0348711}

In[125]:=
```
p[0, r0] - p[0.03487, r0]
```

Out[125]= 26.2745 ← NEW SIDELOBE LEVEL

In[126]:=
```
NIntegrate[f[theta, r0]^2 Sin[theta], {theta, 0, Pi / 2}] * 2 Pi
```

Out[126]= 29.0016 $\propto$ P_{tot} = d integral

In[127]:=
```
f[0, r0]^2
```

Out[127]= 64357.3 $\propto$ P_{max}

In[128]:=
```
NIntegrate[h[r, r0, 37.5] r, {r, 0, 37.5}]
```

Out[128]= 253.688 $\propto$ $\langle E \rangle$

In[129]:=
```
NIntegrate[h[r, r0, 37.5]^2 r, {r, 0, 37.5}]
```

Out[129]= 182.208 $\propto$ $\langle E^2 \rangle$